



Modernizing the chemical engineering curriculum via a student-centered framework that promotes technical, professional, and technology expertise skills: The case of unit operations

Miguel Ángel Ballesteros^a, Juan Sebastián Sánchez^b, Nicolás Ratkovich^a, Juan C. Cruz^{c,d,**}, Luis H. Reyes^{a,*}

^a Grupo de Diseño de Productos y Procesos (GDPP), Department of Chemical and Food Engineering, Universidad de los Andes, Bogotá, Colombia

^b Centro de Innovación en Tecnología y Educación, Faculty of Education, Universidad de los Andes, Bogotá, Colombia

^c Department of Biomedical Engineering, Universidad de los Andes, Bogotá, Colombia

^d School of Chemical Engineering and Advanced Materials, University of Adelaide, Australia

ARTICLE INFO

Article history:

Received 3 August 2020

Received in revised form

18 November 2020

Accepted 22 December 2020

Available online 24 December 2020

Keywords:

Flipped classroom

Professional skills

Information and communications technologies

Unit operations

ABSTRACT

The development of a comprehensive set of skills, including technical, professional, and technology expertise, is critical to succeeding in the increasingly competitive global job marketplace. We proposed to develop such skills in our junior students (third year) via a flipped-classroom approach, a PO-PBL problem, and interactive e-learning tools. The intervention was implemented in the core course of Unit Operations and led to an increase in the students' perception of the development of teamwork and people-related skills. Despite the benefits of promoting student learning, our intervention revealed that we still need to conduct work to approach more robust peer-to-peer interactions and connectedness. In this regard, students showed a marked tendency to have superficial discussions, which reflected their inability to develop superior emotional connections with peers. This is critical to promote complex thinking and ideation as well as continued engagement with the course contents and will be the focus of our future work.

© 2020 Institution of Chemical Engineers. Published by Elsevier B.V. All rights reserved.

1. Introduction

According to the Global Shapers Survey (World Economic Forum, 2017), the most critical problems humanity needs to solve to secure its survival on planet earth are climate change, large-scale wars, religious conflicts, and poverty. These global problems can only be tackled by a critical mass of informed and well-educated citizens with several abilities in formulating and solving real-life challenges. Engineers have been particularly well-suited for supplying novel solutions to some of humanity's biggest challenges throughout history (Wall, 2010). As opposed to earlier concerns, current issues appear multiparametric and complex, where more throughout analyses are needed. Consequently, addressing these issues requires the involvement of professionals from different

disciplines and the participation of various stakeholders from the government, private sector, and non-governmental organizations (Grasso and Burkins, 2010). Accordingly, engineers need to be trained with a more comprehensive set of skills to include competencies in effective communication to different audiences, negotiation, team and resource management, planning and task assignment, use of Information and Communication Technologies (ICTs) and comprehensive data analysis, and the development of lifelong learning strategies (Organisation for Economic Co-operation and Development, 2016; Sein-Echaluce et al., 2018). Despite higher education institutions' efforts to develop the needed skills, only two out of three workers end up having adequate competencies to succeed in the pressing technology-based environment of today's marketplace (Organisation for Economic Co-operation and Development, 2015).

Education innovators have tried to address this challenge by developing disruptive pedagogical schemes such as flipped classroom, project-oriented problem-based learning (PO-PBL), inquiry learning, collaborative learning, digital environments for education, and team-teaching (Hutchings and Quinney, 2015). Most of these approaches are centered on the student, while professors play

* Corresponding author at: Cra 1 N° 18A- 12 Of. ML-744, Bogotá, Zip-code: 111711, Colombia.

** Corresponding author at: North Terrace Campus, Engineering Annex, floor 2, room A207, Adelaide, Zip-code: 5005, Australia.

E-mail addresses: juan.cruz@adelaide.edu.au (J.C. Cruz), lh.reyes@uniandes.edu.co (L.H. Reyes).

a facilitator role to encourage students' interaction with knowledge and peers (Hutchings and Quinney, 2015; Salcedo Galán et al., 2018; Yuen and Hau, 2006). It is important to notice that these approaches were not inherently conceived to develop professional skills exclusively but to improve student learning in broader contexts. Flipped learning is well-suited for engineering education, mainly because engineering students improve their learning processes when they engage in complex problems and projects (Låg and Sæle, 2019; Lombardi and Oblinger, 2007). Moreover, students are more inclined to build and adapt digital learning resources outside the classroom to be prepared before class (Galvis et al., 2017; Salcedo Galán et al., 2018). In contrast, PO-PBL promotes learning by providing students in teams the possibility of addressing a real-life problem (Chen et al., 2020; Jones et al., 2004). Additionally, students learn to manage a project independently by establishing milestones, tasks and roles, resource allocation, and deliverables throughout the project (Kokotsaki et al., 2016). PO-PBL is not only well-suited for the development of functional expertise and technological skill but people-related skills (Lee et al., 2015; Musa et al., 2012).

Under this framework, professors can recognize students' unique needs and interests to make the learning experience as personalized as possible (Crumly et al., 2017; Richards and Burns, 2012). This view was fueled by the constructivism framework put forward by Vygotsky and Piaget (Moore and Piaget, 1971; Vygotsky, 1980). Despite student-centered pedagogical frameworks' success, their initial implementation is hard, mainly due to the paradigm shift involved, as students need to have a more active role. This is also the case of professors as they need to move from mere communicators to providers of robust feedback via both conventional and virtual platforms (Donnelly and Fitzmaurice, 2014; Mergendoller et al., 2015). Flipped classroom innovations in chemical engineering have been recently explored by Falconer et al., where they have confirmed that active learning is a lot more beneficial for students than traditional lecturing in terms of knowledge gain, a better understanding of concepts, engagement in their learning process, and more opportunities for student-professor interactions (Falconer, 2016; Falconer et al., 2012). These studies have also supported the notion that professors are somewhat hesitant to incorporate such innovations as they might require a substantial time commitment. To address this issue, they proposed to develop instructional material and make it readily available for both professors and students outside the classroom. Similar ideas have been put forward by Woods et al. throughout the extensive implementation of active learning classroom setups across the entire chemical engineering curricula. According to this approach, professors need to spend most of the classroom time developing a structured environment to empower students to form teams and learn cooperatively. Simultaneously, such an environment should be positive enough to help students build confidence and process skills, particularly lifelong learning abilities (Estrada et al., 2019; Woods, 2012; Woods and Sheardown, 2009). In this context, lifelong learning is related to each team member's ability to identify key knowledge to address a challenge and come to agreements regarding individuals learning different aspects of what is needed. As a whole, the team should be able to put to practice the knowledge and even elaborate on other potential ways of using it (Vasquez et al., 2020; Woods, 2012; Woods et al., 2013).

An area where engineering curricula have also undergone profound transformations over the past two decades is professional skills development. This has been a response to essential changes in interactions with peers required to succeed within most modern organizations. Nowadays, the complexity of projects requires that engineers are not only capable of mastering and putting to practice technical concepts but respond empathically to forming teams with professionals of different disciplines (Backlund and

Sjunnesson, 2012; King, 2009; Tong, 2003; Woratschek and Lenox, 2002). The development of collaborative strategies, the ability to evaluate peers' performance, and the continuous self-assessment are critical to maintaining a team's performance at the highest levels (Kilic, 2016; Ndoye, 2017). Accordingly, companies such as Apple, Tesla, and Google have developed strategies to maintain healthy team relations and accountability. For instance, in the case of Google, some of these strategies include (i) psychological safety, as team members can take risks without feeling insecure or embarrassed, (ii) dependability, as team members can count on each other to do high-quality work on time, (iii) structure & clarity, as team members have clear goals, roles, and plans, (iv) meaning of work, as team members are confident that work is personally important for them, and finally, (v) impact of work, as team members think that their work matters and create change (Duhigg, 2016; Levasseur, 2017). Psychological safety is, by far, the most critical strategy. It stops people from negatively influencing others through detrimental behaviors such as competition, negativity, and unawareness of others' feelings (Edmondson and Lei, 2014). Under this environment of mutual respect and understanding, team members feel compelled to share their experiences, admit mistakes, form teams, and even explore new roles (Edmondson, 1999; Edmondson et al., 2004; Hinds and Bailey, 2003). Similar strategies have been implemented in the classroom for several chemical engineering courses that revolved around team-taught strategies. In such cases, the interventions' success strongly depended on the students' ability to teach one another and their capability to make sincere and well-thought assessments of their peers (Bosman et al., 2020; Woods et al., 1988, 1997). This was ultimately related to maintaining an environment of trust where asking questions, presenting opinions, taking risks, and others' preferences are well received and encouraged.

The last two decades' internet revolution greatly impacted how higher education is conceived for new generations (Edmunds et al., 2012; Resta and Laferrière, 2007; Saunders and Klemming, 2003; Xu and Shi, 2018). Global access to these platforms has significantly increased over the past few years due to governments' investment in high-speed internet infrastructure and the emerging businesses of mobile internet providers (Bates, 2001; Cambini and Jiang, 2009). As a result, virtual environments and technology-enabled experiences have led to rethinking how concepts are put into practice and competencies are developed (Abidi et al., 2016; Ahn et al., 2015; Alaeddine et al., 2015; Carle et al., 2009; Goldwasser et al., 2016; Kolowich, 2013; Lofstrom and Nevgi, 2008; Wong and Li, 2011). Technological tools have also played a central role in facilitating student-centered pedagogical frameworks (Liberatore et al., 2013). For instance, in PO-PBL, project management can be aided by several emerging technology-enabled tools such as Asana, Planner, Scoro, and Workzone, to mention a few. This continually growing set of platforms offers the possibility of real-time team updates, live chat discussions, and automated reporting. Despite the endless possibilities offered by these platforms, some issues might arise when implemented in real-case scenarios, including access to the internet, technology expertise of team members, user-friendliness of designed activities, and platforms' usability (Leask, 2012; Melor et al., 2009). Similar ICT tools have been implemented to increase student engagement during the term, directly reflected in facilitating their lifelong apprehension of concepts. For instance, according to Liberatore et al., when students develop in-class or homework problems with the aid of rich multimedia materials (e.g., YouTube videos), the vast majority of them came out with rigorous solutions that demonstrated a much more profound understanding of topics when compared with a traditional textbook approach to solving problems (Liberatore et al., 2013). Moreover, recent reports have concurred that a robust strategy is mixing active learning with digital tools to promote learning of the foundational disciplinary

knowledge of chemical engineering and more general transferable skills (Hasna, 2010; Mio et al., 2019; Rashid and Ventura-Medina, 2012). In this regard, an emerging trend is to seek students' active engagement by deploying scenario-based learning tools online (Mio et al., 2019). This approach relies on bringing students in close contact with real industry problems and ask them to address such challenges by drawing on previous knowledge and putting to practice concepts.

This work is dedicated to evaluating whether a proposed framework for teaching the junior (third year) Chemical Engineering core course - Unit Operations (UnOps) - is suitable for developing a set of crucial skills currently on-demand in the practice of the profession. The developed framework puts forward the idea that the needed technical skills, professional skills, and technology expertise can be successfully exploited by properly combining three emerging disruptive pedagogical methodologies: flipped classroom, ICT, and project-oriented problem-based learning (PO-PBL). The course was tailored such that each of these components holds a significant focus on the development of the needed skills. We wanted to evaluate whether the proposed learning materials to support the flipped classroom approach was varied, interactive, flexible, clear, and resourceful enough for the students. Additionally, we wanted to see if the technology tools promoted positive interactions and more profound and independent reflections on the course topics. Moreover, we were interested in determining the role of rich peer-to-peer communication and participative environment on the fulfillment of learning goals.

2. Course redesign: unit operations

The starting point of the study was that previous surveys to students showed that they perceived the course mentioned as highly demanding and located in a semester along with other two courses with similar difficulty and time commitment (i.e., Unit Operations, Separation Processes, and Introduction to Modeling and Simulation). Therefore, the associated academic load triggers a significant sense of aversion towards enrolling in the three courses simultaneously. To address this issue, the techno-pedagogical reforming team, i.e., Conecta-TE (Center for Innovation in Technology and Education), the professors in charge, and the graduate teaching assistants, proposed an initial intervention back in 2017–1. This first approach was oriented towards developing a joint project (PO-PBL approach) for Unit Operations and Introduction to Modeling and Simulation (Ballesteros et al., 2019). According to these previous results, students perceived the joint project as positive, and 85 % of them appreciated the design component of it, considering that it benefits their learning process. The challenge it presented provided them with new tools in managing chemical engineering problems. The students were challenged to make well-informed and conscious decisions, which is a skill that would benefit their professional and personal future. Even though the PO-PBL approach provided the students with a set of professional skills for teamwork and project management, both the main technical concepts and the use of modern ICT tools lag in development. This lag in development was considered worrisome as these two aspects are central to a chemical engineer's professional practice.

In the case of technical skills, these fundamental concepts were the transport of liquids, transport of gases, stirring and mixing, heat exchangers, solids handling, solids and fluids storage, and engineering diagrams. These are shown as the light blue gears in Fig. 1A. The ICTs are depicted as an external framework shown in golden in Fig. 1A. Multiple tools support each of the technical concepts and promote professional skills such as team communication and autonomy. Finally, the gray blocks in Fig. 1A are for the foundation of an environment of trust and study habits, which could be

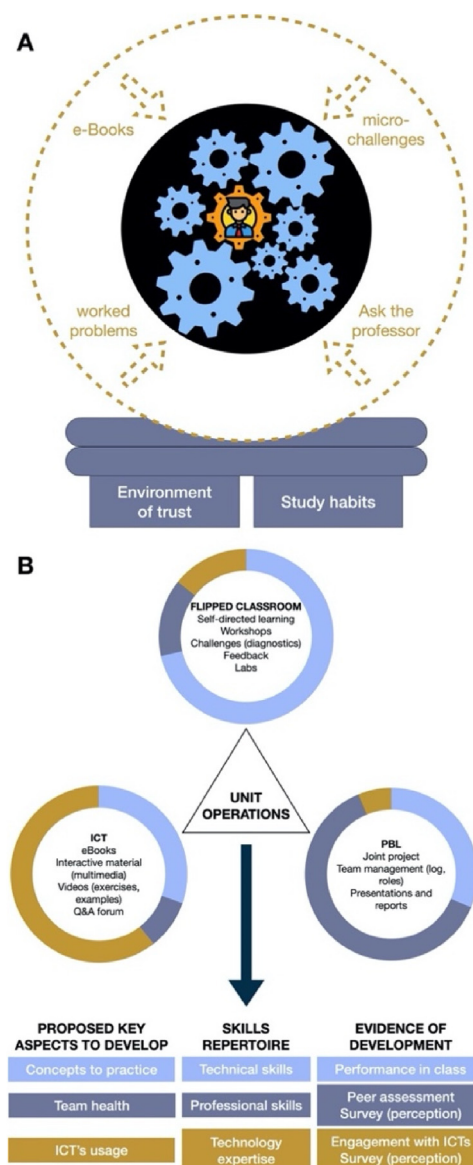


Fig. 1. A. Student-centered view of the proposed techno-pedagogical approach. Light blue gears are for the seven core technical concepts of the course. The golden dotted line stands for the ICT framework supplied to enrich the course enrollees' e-learning experience. The gray blocks depict the foundation of student learning supported on study habits and an environment of trust, correlated to people-related skills and teamwork abilities. B. The professor's view of the redesigned UnOps course, where the flipped classroom and ICT components were built upon the PO-PBL approach, developed on a first intervention (published data) (Ballesteros et al., 2019). The corresponding circular sector's size represents the relative contribution of each type of skill within the proposed framework. Evidence of skills development was gathered from surveys, peer assessment, and students' overall performance (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

correlated with how students assume roles in teams and communicate with their peers. This holistic perspective was the basis for the proposed student-centered, techno-pedagogical approach that served to redesign the course. Fig. 1B shows the professor's view of the newly design UnOps course. Therefore, the idea was to build the two extra components, i.e., ICTs and flipped classroom, upon the already successful PO-PBL framework (Ballesteros et al., 2019). This was with the ultimate goal of providing a more balanced set of tools to develop a skills repertoire that genuinely reflects the notion put forward by the School, of a well-rounded and integral chemical engineer capable of responding to the modern challenges

of the profession. The corresponding relative weights of each of the developed skills per component are schematically shown as fractions of the circular sectors in Fig. 1B. To assess whether the skills were effectively developed, the results of surveys, peer assessment, and overall performance were thoroughly analyzed.

In summary, the proposed intervention was oriented towards much more comprehensive development of a repertoire of skills for our junior undergraduate students in Chemical Engineering. This was accomplished by a flipped-classroom approach where new and varied technology tools were introduced to promote student autonomy and a more productive class environment where students had higher chances to openly approach the teaching team without the commonly found communication barriers. Additionally, we were committed to exploring whether new routes for curricular development that differentiate from traditional views based on misconceptions regarding the role of learning styles.

2.1. Curriculum design to strengthen technical skills

The first step towards an overhaul of the course was to formulate the course learning goals and develop the competencies in the students (Table 1). These were determined based on the entry profile (described in the Methodology section) and the chemical engineer's needs in its professional life. The course contents were established based on the fundamental concepts previously mentioned, creating seven modules of study (Fig. 2A). Even though the modules can be arranged differently, the illustrated sequence was selected to ease the integration with the course Introduction to Modeling and Simulation. Also, it aided the selection of joint projects and connections with later courses such as Process Plant Design (the capstone project).

In line with the development of technical skills, the UnOps course introduced a flipped-classroom approach, which was intended to promote the apprehension of concepts via out-of-class activities. The goal was to increase the readiness of the students before lectures such that professors could focus on providing feedback on concepts representing a significant challenge for students. The hypothesis was that professors could use time more effectively as students prepare the class topics in advance. Accordingly, three learning instances were used during the course: BEFORE CLASS, IN CLASS, and AFTER CLASS, as shown in Fig. 2B. The BEFORE CLASS instance includes activities corresponding to preparatory reading, micro-challenges, an e-space for professor feedback, and worked problems via ICT tools (Supplementary Table 1). The second instance was IN CLASS. Initially, the feedback was carried out to clarify doubts regarding the material examined and analyzed independently. This activity was conducted with the professor promoting active participation to address questions cooperatively using short one-on-one or team discussions. He mediated knowledge acquisition by encouraging peer-to-peer discussions, problem-solving strategies, and delving into the studied topics and examples of various worked problems. Likewise, a series of periodic formative workshops were assigned, thereby allowing them to understand and develop skills in putting concepts-to practice. The resulting doubts or difficulties were then addressed IN CLASS. Finally, different AFTER CLASS activities were planned, such as completing the IN CLASS formative workshops and developing two joint projects throughout the term (in conjunction with Introduction to Modeling and Simulation). The projects were designed so that students were confronted with a challenge that approached those commonly found in chemical engineering's professional practice. Furthermore, students conducted lab experiments to strengthen concepts and develop activities supporting the projects' experimental schemes.

2.2. Curriculum design to strengthen professional skills

Even though the first pedagogical intervention introduced the joint team project, its focus was on developing technical skills. As a result, a higher emphasis was given to developing the team skills for the later interventions. Herein, professional skills were, therefore, implemented incrementally along with the intervention (Supplementary Table 2). Since its conception, we introduced a joint project with the Modeling and Simulation course. The results of such intervention were described in our previous paper. The peer-to-peer evaluations (co and auto-evaluations) were then implemented in 2017–20, which provided an opportunity for team members to present their perceptions of overall team performance and how others see their involvement in the planned activities (see Supplementary Data to consult the corresponding format). Team conflicts continue to be perceived as an issue, which motivated us to include, during the same term, a Team Contract where students, from the very beginning of the semester, agree upon their roles and time commitment to assure a full and timely completion of projects. This was a crucial milestone of our intervention as evidenced by more and sustained peer-to-peer interactions and ultimately was reflected on the quality of the projects. Finally, to have more accountability and extra negotiation routes to deal with possible conflicts, we implemented follow-up minutes in 2018–20 where students had the chance to address emerging issues rapidly, and the professors and teaching assistants have an easily-accessible repository with a recount of team events. This was crucial to develop a much more robust and comprehensive team intervention strategies. Additionally, professional skills in the flipped classroom and ICT components were designed to potentiate individual habits such as self-regulation and autonomy.

2.3. Curriculum design to strengthen technology expertise

The designed flipped classroom strategy was enabled by a robust set of e-learning and multimedia tools, including e-books, interactive presentations, YouTube example video clips for each module, reading verification micro-challenges, and supplementary worked problems. The e-books were pdf files containing the technical foundations for the unit operations studied during the course. In interactive presentations, they were developed in HTML5 embedded into the online platform SICUA+ (Blackboard). The micro-challenges were set up to aid the tools available in the online platform SICUA+ (Blackboard Inc., Washington D.C., USA). They consisted of eight multiple-choice questions randomly selected from a database of about 40 questions per weekly topic. Likewise, an online space called "Ask the professor" was conceptualized and implemented as a forum where students ask questions addressed by the professor at the beginning of the following class. All new material developed for the course can be accessed from the links in Supplementary Table 1 and a screenshot of the SICUA+ platform, and the e-books (Supplementary Fig. 1). Additionally, the ICTs were incrementally implemented along with the duration of the intervention. As shown in Supplementary Table 3, the initial intervention occurred in 2017–20, after the entry profile survey. This intervention consisted of solids handling resources, specifically the solids and size reduction interactive presentations and the corresponding eBook chapters. Feedback from this iteration showed insufficient worked problems for student's self-directing studies. This led to creating the YouTube Example Videoclips and devising the forum Ask the professor to give first-hand feedback in 2018–10. During this semester, the eBook was expanded to cover all the seven modules described in Fig. 2A. Students' perceptions from 2018–10 uncovered some deficiencies in the worked problems, as the students described them as too simple. Supplementary learning examples with a higher level of difficulty were created for each

Table 1
Objective and technical, professional, and technology expertise competences of the redesigned course.

REDESIGNED COURSE
LEARNING GOALS:
- To select, size, and design industrial equipment for the transport and handling of fluids (pipes, turbomachines, and heat exchangers) and solids handling (size reduction systems), to be used later in the process design, applying concepts seen in the courses of Industrial Processes Fundamentals and Transport Phenomena. - To facilitate peer-to-peer interactions so that students can propose impactful teamwork strategies and assume roles according to their habits and learning styles. - To integrate ICT tools into the students' study strategies to facilitate the accomplishment of learning goals.
COMPETENCIES TO DEVELOP WITH THE COURSE:
- Technical skills. - To determine (calculate) the dimensions of industrial pieces of equipment for particular unit operations and in the context of an application. - To apply design criteria to select the appropriate equipment for specific unit operation. - To evaluate different design options in the transport and handling of fluids and solids. - To elaborate diagrams with information that reflect proper engineering design and a communicative intention for different audiences. - To propose key performance indicators related to the fulfillment of the objectives of a project. - Professional Skills. - To negotiate with peers about the involvement in tasks, playing roles, and time commitments through a project. - To play an active role in a project after a conscious self-understanding of strengths and weaknesses. - To properly allocate resources to teams working on solving problems associated with an industrial process's general operation. - To make rigorous both self- and peer-assessment processes regarding habits and their impact on a team's performance. - To plan the timeline for the execution of a project under resource and timeframe constraints. - Technology expertise. - To use e-learning and rich multimedia tools to facilitate learning processes. - To reflect upon selecting and incorporating the appropriate e-learning and rich multimedia tools into study habits.

Table 2
Instruments used to assess course perception along with the redesign of the UnOps course.

Activity	2017-1: context	2017-2: module pilot	2018-1: course pilot	2018-2: implementation
Professor interview (formative)	X			
Entry profile (formative)	X			
Focus group 1 (formative)		X	X	
Focus group 2 (formative)			X	
CCAA survey (formative)	X	X	X	X
Effects and impacts survey (summative)		X	X	X

of the modules described previously. Additionally, the eBook was expanded with topics that are not typically covered in the course, such as Drying and Sedimentation.

3. Methodology

The pedagogical intervention to the UnOps course was carried out during four consecutive academic semesters: 2017-1, 2017-2, 2018-1, and 2018-2, and it was tracked using both formative and summative evaluations (Table 2). In the formative assessment case, we included both qualitative (from focus groups) and quantitative information conducted by studying the *fundamental conditions of the learning environment* (CCAA survey, see below for detailed description). Additionally, a formative diagnosis was also conducted only in 2017-1 to evaluate teaching and learning perceptions *a priori* with the aid of (i) a professor interview inquiring about teaching practices, (ii) an entry profile survey (100 % of enrollees, see below for detailed description), and (iii) the CCAA survey (81.5 % of enrollees). The formative evaluation was used to provide continuous feedback and improvement to the course design and implementation. At the same time, the summative evaluation (effects and impacts survey), which only included quantitative information, was dedicated to evaluating the learning experience according to the proposed objectives.

During 2017-2, the first set of redesigned activities was introduced for one of the main modules of UnOps (Solids Handling) that included all ICT tools and the first approach to a flipped classroom. To evaluate whether the proposed intervention was according to the students' expectations, only one focus group was assembled. The proposed intervention was fully implemented in 2018-1, and by the end of the term, two focus groups were interviewed. As opposed to 2018-1, where the professors had the support of the

Center for Innovation in Technology and Education (Centro de Innovación en Tecnología y Educación - Conecta-TE), for the 2018-2 term, the course team conducted an utterly independent implementation.

The sample size and demographics of the UnOps enrollees for this study are presented in Supplementary Table 3.

3.1. Entry profile survey

The entry profile survey was applied before designing the intervention, i.e., 2017-1. The number of respondents was 38 people. The entry profile survey includes a Likert scale from 1 (strongly disagree) to 5 (strongly agree) to build the index of the proportion of students (Eq. 1) where x_1 stands for "strongly disagree," x_2 for "disagree," x_3 for "neutral," and x_4 for "agree." Finally, x_5 indicated the student "strongly agree" on the subject. The variable x represents the fraction of responses in the subset.

$$\text{Index}(\text{Question } n) = x_1 + 2x_2 + 3x_3 + 4x_4 + 5x_5 \quad (1)$$

Supplementary Table 4 shows a detailed description of the surveyed topics in each of the three main skill categories: technical, professional, and technology expertise. This information is provided for the entry profile, CCAA, and the Effects and Impacts surveys. The reader is encouraged to review the surveys' topics before continuing to the data analysis section. The entry profile survey was designed to look at students' perceptions of technical skills, particularly their notions of putting concepts to practice and academic load. This was also the case for the professional skills that revolved around teamwork, motivation, and habits. For technology expertise, the focus was on the students' ability to interact with ICTs. The ICTs in education allow the interaction between stu-

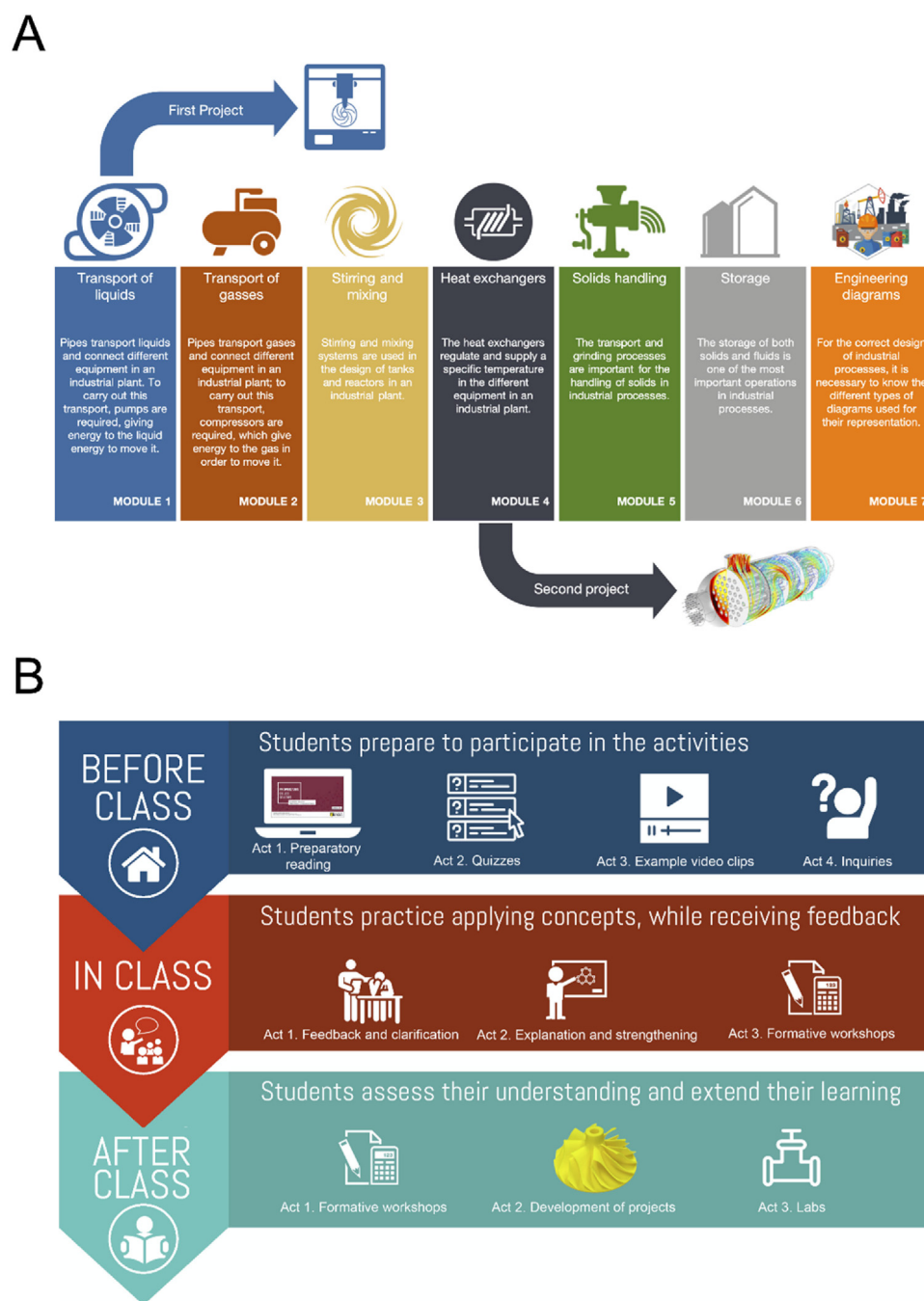


Fig. 2. A. Course contents for the redesigned course. The main concepts were distributed into seven sequential but interchangeable modules that take place during the term. B. Course methodology using the flipped classroom approach. Activities BEFORE CLASS included preparatory reading, quizzes, and video clips. Activities IN CLASS included feedback and clarification and strengthening. The events AFTER CLASS included projects and lab practices. Formative workshops were both for IN CLASS and AFTER CLASS activities as they were often started during classes and complete afterward.

dents, thereby facilitating teamwork. Therefore, the attitudes of students toward the use of such technologies require a comprehensive analysis. For this purpose, the user-friendliness of the learning managing system (SICUA+) was analyzed during the term. Regarding the professional skills category, three aspects were surveyed: the motivation for learning, teamwork, and self-regulated learning.

3.2. Fundamental conditions of the learning environment (CCAA) survey

The CCAA is an appealing tool to scrutinize environmental and psychological learning contexts, different resources, classroom

restrictions, and strategies used to promote students' learning (Galvis, 1992). This survey examined the students about the three previously mentioned main skill categories: technical, professional, and technology expertise (Supplementary Table 4). The CCAA survey was consistently applied during the intervention, first as a diagnostic tool (2017-1), and then as a continuous improvement one (2017-2, 2018-1, and 2018-2). As for the entry profile survey, the CCAA survey followed the Likert scale (Eq. (1)) and focused on the students' progress in the three main skill categories. Additionally, the study included open questions to gather further qualitative information about their perceptions of the designed framework (Supplementary Table 4). Concerning the diagnostic entry profile,

the CCAA provided a more comprehensive and in-depth interrogation for each skill.

3.3. Effects and impacts survey

The effects and impacts instrument also used a Likert scale to create an index of proportions to quantify the course redesign's impact on the students' learning experience, and the categories scrutinized are shown in Supplementary Table 4. It is important to emphasize that this survey was used as a summative assessment, where the results are communicated only at the end of the semester when there is no longer the possibility of adjusting and focused on determining if the objectives set at the beginning were reached.

3.4. Statistical analysis

Statistical significance was evaluated using a two-way ANOVA considering continuous values of the Likert scale obtained from the surveys by employing the software GraphPad Prism version 8 for MacOS (GraphPad Software, San Diego, California, USA), where (*) corresponds to a p-value < 0.05, (**) to a p-value < 0.01, (***) to a p-value < 0.001, and (****) to a p-value < 0.0001.

3.5. Ethics statement

The institutional review ethics board approved this research and granted permission for human subjects in this study; written consent was obtained from all participants (Supplementary Data).

4. Results

4.1. Entry profile survey and interviews with focus groups and professors

About half of students took the course later on their study plan, which was attributed to the initially planned semester's excessive academic load (i.e., the end of the Junior year). In this regard, 91 % of students described the load as either high or very high. This suggests that even though the course was initially designed for a load of 3 credits (i.e., three h of work in class plus six h of individual work per week for a total of 144 h during the semester), the real required time commitment went well beyond that. The survey revealed that students have a positive attitude towards using ICTs, as evidenced by evaluations of 4.00 and above (Supplementary Table 5). Focus groups were assembled to further interrogate whether the survey findings were robust enough to use them for the course redesign. This was accomplished by conducting interviews with both the focus groups and the course professors. When asked about motivation, most students (91.67 %) stated that they only take the course because it is mandatory. Only 33 % said to have a real interest in the course topics. This was in line with the findings of the entry profile survey. About 40 % of the interviewed students communicated that the previous knowledge is insufficient, and particularly some of the topics covered in the courses transport phenomena (fluid mechanics and heat transfer) and mass and energy balances.

4.2. Techno-pedagogical intervention

The entry profile survey and focus groups confirmed that the initial intervention was incomplete and only promoted a partial development of professional and technical skills (Ballesteros et al., 2019). Therefore, a significant intervention was needed to propose more effective concepts-to-practice strategies, strengthen teamwork and people-related abilities, and incorporate state-of-the-art ICTs to enhance students' learning experiences.

In the case of professional skills, the intervention was oriented to targeting teamwork skills that included positive interdependence and collaboration, negotiation, resource management and planning, and peer- and self-assessment of habits that might impact a team's performance. A detailed description of each of the evaluated aspects within the Effects and Impacts Survey for each of the skills components are shown in Supplementary Table 6. In Fig. 3, the scores for technical skills (TS_{EI}), professional skills (PS_{EI}), and technology expertise (TE_{EI}) for the Effects and Impacts Survey can be found. Fig. 3A shows the time evolution of scores for items with significant changes over time for the technical skills part and average values in the same time frame. The results confirmed that, on average, the students' perception of the development of technical skills significantly improved from 2018-1 to 2018-2. Similar results were obtained for the professional skill component (Fig. 3B). These findings can be explained by several improvements in the course structure and the implementation of new learning tools during 2017-2. In the case of technology expertise, significant differences were found between 2017-2 and 2018-2, which can be attributed to higher scores for 2018-1 than those obtained for the other two components (Fig. 3C). A comparison of average changes in scores during the observation period for the three components is shown in Fig. 3D.

Fig. 3A shows that for the item TS2_{EI} of technical skills, "This course has allowed me to choose among a variety of learning materials," the score also increased from 2018-1 to 2018-2. This reflects that the students perceived that several tools could enable learning and that such tools can vary according to different learning styles. Fig. 3B shows similar results for PS11_{EI}: "The course promotes an environment for collaboration with peers" and PS15_{EI}: "This course offers the opportunity to rely on peers when my understanding of a particular topic is limited." Items with extensive data dispersions were selected as targets for further improvement. Some of the items from the technical skills component where opportunities may arise in the future include TS13_{EI} "The interaction with the resources of this course motivated me to consult other sources of information" and TS4_{EI} "The methodology of this course adapts to my learning needs." One item from the technology expertise component that might be subject to future intervention is TE10_{EI} "In this course, I received timely feedback from my teacher."

A detailed description of each of the evaluated aspects within the CCAA Survey for each of the skills components is shown in Supplementary Table 7. In Fig. 4, the scores for technical skills (TS_{CCAA}), professional skills (PS_{CCAA}), and technology expertise (TE_{CCAA}) for the CCAA Survey can be found. Fig. 4A shows the time evolution of scores for items with significant changes over time for the technical skills element and average values in the same period. The results confirmed that, on average, the students' perception of the development of technical skills significantly improved from 2018-1 to 2018-2. Comparable results were obtained for professional skills (Fig. 4B) and technology expertise components (Fig. 4C). A comparison of average changes in scores during the observation period for the three components is shown in Fig. 4D.

Fig. 4A shows that for the items TS1_{CCAA} "I received a satisfactory answer to resolve my concerns about the course activities," TS6_{CCAA} "The interactive resources were relevant to my learning," and TS9_{CCAA} "The information contained in the interactive resources was sufficient," the score significantly increased from 2018-1 to 2018-2.

In the case of professional skills (Fig. 4B), significant differences were found only for PS1_{CCAA} "The distribution of activities over time was adequate" and PS9_{CCAA} "In this course, peer feedback was promoted." For the technology expertise component, the only item where significant changes were observed was TE13_{CCAA} "The information contained in the eBooks was clear" (Fig. 4C). Along with the other ICT tools and activities designed for better student experience, the eBooks provided a route to access knowledge that is

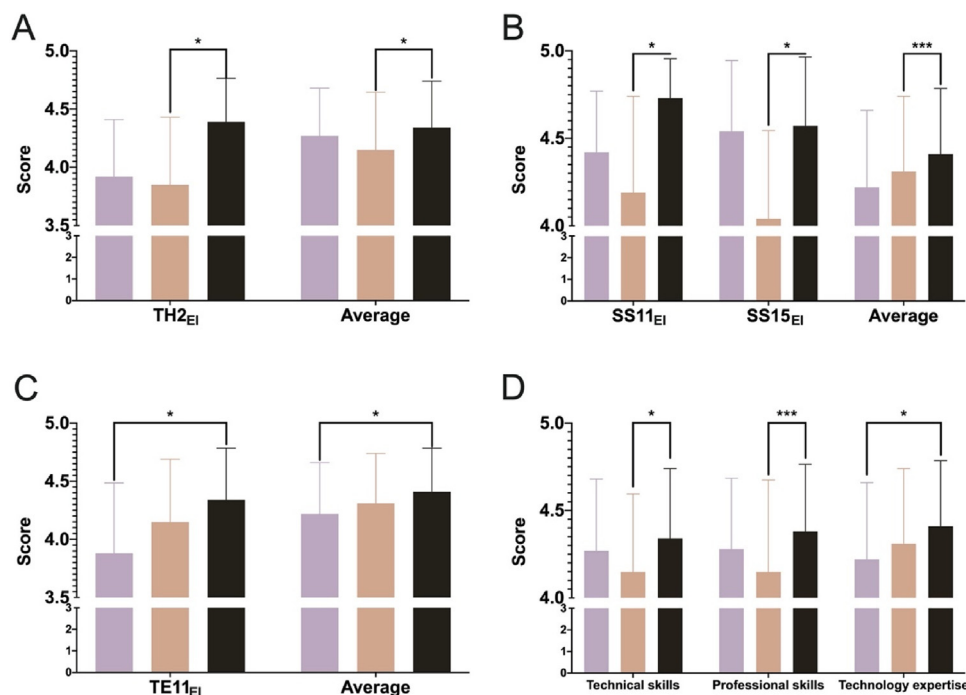


Fig. 3. Effects and impacts scores 2017-2 (violet), 2018-1 (brown) and 2018-2 (black). A. Scores for technical skills (TS_{EI}). B. Scores for professional skills (PS_{EI}). C. Scores for technology expertise (TE_{EI}). D. A comparison of average changes in scores during the period of observation for the three components. A two-way ANOVA was performed to assess significance, where (*) corresponds to a p-value < 0.05, and (***) corresponds to a p-value < 0.001. The description of the categories can be found in Supplementary Table 6. For this study the number of respondents was 21 in 2017-2, 46 in 2018-1 and 37 in 2018-2.

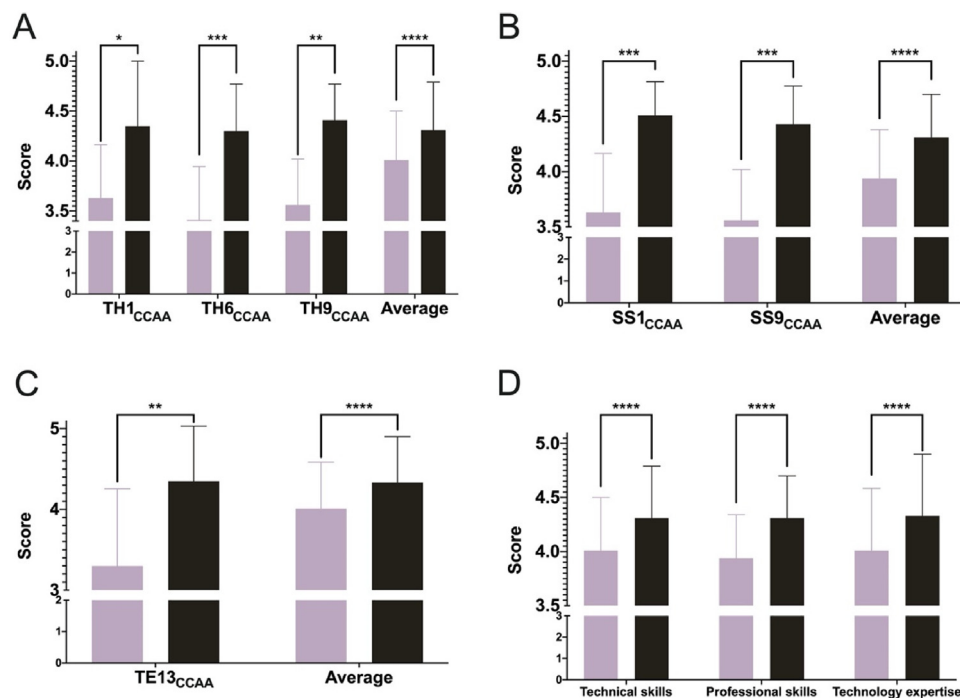


Fig. 4. Results of the CCAA survey scores 2018-1 (violet) and 2018-2 (black). A. Scores for technology expertise (TS_{CCAA}). B. Scores for professional skills (PS_{CCAA}). C. Scores for technology expertise (TE_{CCAA}). D. A comparison of average changes in scores during the period of observation for the three components. A two-way ANOVA was performed to assess significance, where (*) corresponds to a p-value < 0.05, (**) to a p-value < 0.01, (***) to a p-value < 0.001, and (****) to a p-value < 0.0001. The description of the categories can be found in Supplementary Table 7. For this study the number of respondents was 46 in 2018-1 and 37 in 2018-2.

appealing for the learners (for details on the eBook, please refer to Supplementary Table 1). Further evidence of students' superior engagement with the designed ICTs is provided by the results of views of YouTube video clips (Fig. 5C). It was noted that close to the end of the 2017-2 semester, video clip views approached seven on average per team (two students) with an average engagement

time of about 4 min per group, which is the video's duration. For the 2018-1 semester, six clear peaks of engagement were observed during the term. These peaks correspond to the six videos designed for the students' enhanced ICT experience (Supplementary Table 1). The first two peaks corresponded to the pipelines and pumps modules and were intended to support the first homework and

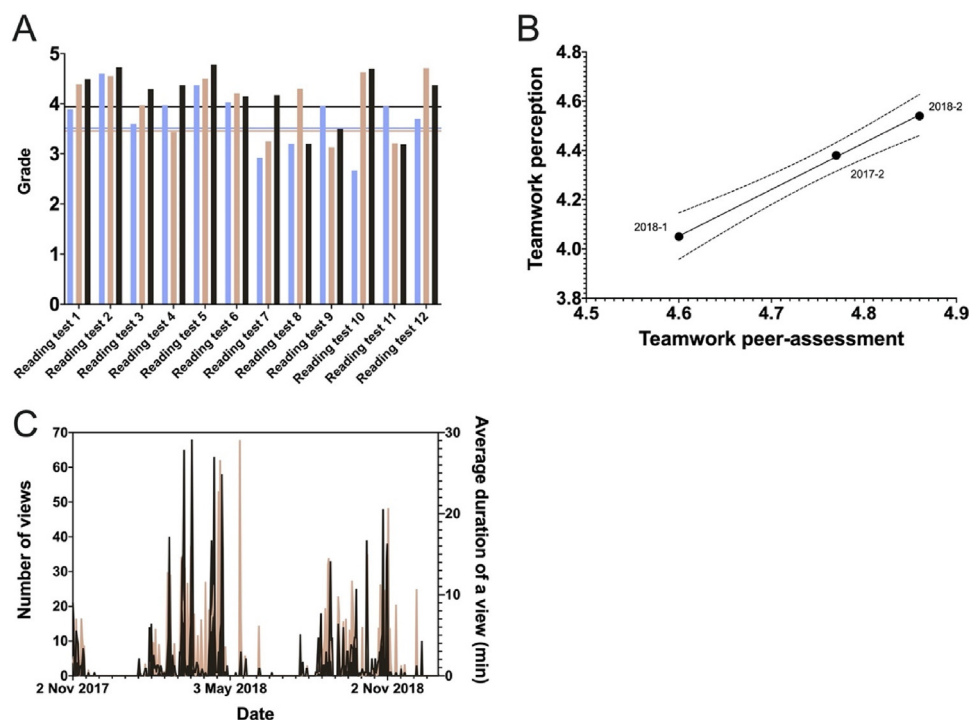


Fig. 5. Evidence of development for technical, professional, and technology expertise skills. A. The average grade of the micro-challenges during the implementation. 2018–10 (violet) and 2018–20 (brown). The semester 2019–10 (black), although not part of the study, was added to show the continuous improvement of grades. The horizontal lines show the final grade obtained in the course. B. Comparison between the teamwork peer-assessment and its perception using the surveys. Dotted lines represent the 95 % confidence interval of the slope. C. The number of views (black) and the average duration of a view (brown) of the YouTube video clips during the implementation.

exam preparation. In this case, average views approached 5 per team with an average duration of about two and a half minutes. The second set of peaks was for the compressors and stirring and mixing modules included in the second homework and exam. For this latter group, views per team increased to 9 with an average engagement time of about 5 min. The final set of peaks was for the solids and heat exchangers modules, and in this case, views increased to 11 and engagement time to 6 and a half minutes. Similar findings were obtained for 2018–2, where engagement and views increased in time. As for the Effects and Impacts Survey, some of the items might be suitable for further improvement. The selected items were those with the most significant data dispersions. In the case of technical skills, it is still possible to continue improving TS1_{CAA} “I received a satisfactory answer to resolve my concerns about the course activities.” The professional skill component might be subject to improvement in PS6_{CAA} “The technological tools of the course facilitated my interaction with other students,” PS14_{CAA} “The monitors created a trustful environment for students to participate and ask questions,” and PS15_{CAA} “The graduate assistant created an environment of trust for students to participate and ask questions.” The last group of items for improvement belongs to the technology expertise component. It includes TE8_{CAA} “The contents were available on the platform promptly,” TE9_{CAA} “I received a timely response to resolve my concerns about the course activities,” and TE10_{CAA} “I received a timely response when I requested support to resolve technological difficulties.”

5. Discussion

5.1. Entry profile survey and interviews with focus groups and professors

According to the students’ perceptions, ICTs are valuable to improve the learning processes during their careers, thereby mak-

ing the topics more exciting and the activities more productive. Additionally, they emphasized that ICTs might promote lifelong learning of critical concepts in chemical engineering. This aligns well with the findings of Woods et al. on the implementation of PBL, according to which this long-term process of knowledge apprehension is related to a well-thought problem-solving strategy (Woods, 2012; Woods et al., 1997).

Despite students’ interest in a deeper understanding of the course contents, their most important driver to get involved in the course activities was the performance in grades followed by lifelong learning. This was confirmed by only 33 % of enrollees showing a genuine interest in putting to practice the course topics. Competition with peers, the possibility of materializing creative ideas, and personal motives were disregarded in importance as drivers. The self-regulated learning category (Supplementary Table 5) shows that students tend to have a passive involvement in intervening in their habits to move to better learning processes. This is evidenced by their awareness of the role they play in the learning process but also by the difficulties in defining and implementing strategies for continuous self-assessment and routine activities. Consequently, this is an area of further intervention, as suggested by recent reports that advise professors to find new methods to construct knowledge through self-regulated learning (Eom and Ashill, 2018; Michalsky and Schechter, 2013; Ng, 2018). Moreover, it has been shown that self-regulated learning skills should be promoted as they are vital to plan and set goals during projects and ultimately ensure their timely completion. This, in turn, is also important to evaluate tasks and activate prior knowledge (Doo and Bonk, 2020).

Teamwork was identified by students as necessary and perceived that their involvement is essential. Also, they believe that their peer opinions are respectable. In contradiction to this view, however, students have difficulties trusting others’ work and find routes to identify and address teamwork issues. This might be

related to limited involvement, derived from students' psychological need to search for sufficient relatedness with peers, i.e., the ability to extend genuine emotional bonds with others (Leenknecht et al., 2017). A limited involvement environment has been shown to detrimentally impact peer feedback as this might be mediated by several affective features such as the sense of belonging, the level of acceptance, and anxiety (Mercader et al., 2020).

The focus groups revealed “knowledge gaps” concerning previously acquired knowledge, which the professor also mentioned during the interview. This finding supports the notion that the academic load is unbalanced for the UnOps course as extra time needs to be invested in reviewing key topics that should have been previously covered. Students confirmed that the development of routines and study habits is challenging for them. For this reason, the professor reflected that the course contents should be more related to real problems supported by IN CLASS practical activities such that students increase their level of engagement and attention.

5.2. Techno-pedagogical intervention: assessment of skills components from the effects and impact survey

The results indicate that the items PS11_{EI}: *The course promotes an environment for collaboration with peers* and PS15_{EI}: *This course offers the opportunity to rely on peers when my understanding of a particular topic is limited* showed an increased perception from 2018-1 to 2018-2. This trend suggests that students feel comfortable working in teams and trusting team members to fill in knowledge gaps. This aligns well with the idea put forward above, according to which, as time passed, the environment was sufficient for peer-to-peer interactions at a superficial level. Such interactions appear sufficient for rapid ideation, solving simple challenges, and collaboration in putting to practice readily accessible concepts. However, higher-level thinking might be challenging to achieve as teams largely failed to connect at a deeper level (King, 2002; Wentzel and Watkins, 2011). This notion has been supported by numerous recent reports that indicate that students' learning processes are primarily benefited by regular social interactions with peers, teachers, and experts (Burgess et al., 2019; Puntambekar, 2006). According to these studies, successful team learning not only requires that members share an interest in gaining proficiency in a subject but a common ground concerning shared language, resources, concepts, experiences and tools (Handley et al., 2006).

Most importantly, they must engage in joint activities and interactions to ensure long-lasting learning relationships (Burgess et al., 2019). As described by Menekse et al., such high-quality interactions ultimately translate into superior individual learning gains. This is particularly important for teams involved in constructive activities such as formulating questions, comparing and contrasting cases, integrating text and diagrams, drawing analogies, and generating predictions (Menekse and Chi, 2019). Moreover, it appears that the quality of interaction is mediated by certain types of discourse moves (e.g., elaboration) and dialogue patterns (Menekse and Chi, 2019). As a result, novel approaches oriented towards developing learning communities with team-based learning have become popular for STEM students (Haidet et al., 2014; Parappilly et al., 2019).

In the case of technology expertise (Fig. 3C), significant differences were found for TE11_{EI}: *Resources available for this course promote reflection about some of the covered topics*. This result agrees well with the students' perception that new tools were made as readily available and useful sources to consult and put the main concepts to practice. Moreover, it aligns well with recent reports according to which learning in a technology-enabled active learning environment leads to superior benefits to students compared with the traditional lecture-based environment (MacLeod et al.,

2019; Shi et al., 2020). Despite the growing interest in such methodologies, this has impacted different aspects of student behavior, including the time, place, path, and pace at which students learn (MacLeod et al., 2018). Also, because such scenarios tend to discourage face-to-face interactions, it is of the utmost importance to promote positive student-to-student connectedness. This suggests that technology-mediated instructional approaches' success requires enhanced routes for social interaction and communication between peers (Gong et al., 2020; MacLeod et al., 2019; Sidelinger et al., 2015). Studies regarding students' preferences during such interventions are relatively scarce. However, our contribution provides initial evidence for the notion that the development of metacognitive skills and a self-assessment of learning needs is encouraged by the use of ICTs (Hollingworth and McLoughlin, 2001; Valentin et al., 2013; Voogt and Pelgrum, 2005; Vovides et al., 2007). No other items exhibited significant differences during the intervention period, which implies that there is still space for further improvement. We even found that the item TE10_{EI}: *In this course, I received timely feedback from my teacher* showed a decline during the same period. This suggests that the professor needs to improve some strategies to respond to students more attentively. This can be possibly accomplished by implementing tools that can be handled at the fingertip on a mobile device. Mobile learning has been recently described as significantly successful in positively creating psychological constructs responsible for enhancing learning processes (Bernacki et al., 2020).

Moreover, students have found mobile platforms useful to enact learning strategies, seek help, and engage in computer-supported collaborative learning (Hsu and Ching, 2013; Jeng et al., 2010; Reeves and Sperling, 2015). This is supported by recent works that demonstrated that when feedback is provided promptly, students tend to take better advantage of it and view it as a dialogue with educators or peers (Boud and Soler, 2016; Lam, 2017). In conclusion, much effort will be then needed to elucidate whether, in the context of the UnOps course, the proposed ICT framework is well-suited to promote sufficient classroom connectedness, out-of-class communication (OCC), and out-of-class self-regulated and peer learning.

Finally, we identified two items for improvement within the technical framework, namely, TS13_{EI}: *The interaction with the resources of this course motivated me to consult other sources of information* and TS4_{EI}: *The methodology of this course adapts to my learning needs*. The first item corroborates the notion that even though ICTs provide an alternative to promote engagement, it might be superficial where mainly cognitive constructs and processes are considered. In contrast, the emotional and behavioral ones are left out (Fredricks et al., 2006). For this reason, beyond providing a rich platform for exploring activities at will, the ICT framework should encompass tools that motivate students to connect at a deeper level with peers and instructors. Examples of such tools include concept maps (Bauman, 2018), and virtual agents and multimedia animations (Song et al., 2019), as they can be built for cooperative interactions. This approach has been reported to help controlling anxiety and boredom while promoting passion and a more profound interest (Henrie et al., 2015). The second item for improvement provides further evidence for the need for much more comprehensive tools. A recent example of an innovative approach for superior is the Academic Communities of Engagement (ACE), which not only considers more types of learning activities but the role of out-of-class interactions through extended personal support communities (i.e., family and friends) (Borup et al., 2020).

5.3. Techno-pedagogical intervention: assessment of skills components from the CCAA survey

We attributed the improvement in the overall perception from the pilot in 2018-1 to the full implementation in 2018-2 to the

gained experience of the teaching team. In this regard, we believe that professors were considerably more confident and assertive in customizing the application of the developed tools according to the students' perceptions.

TS1_{CCAA} showed an increase in score for the period of analysis, which appears to reflect that learning was promoted by an increased commitment of professors and teaching assistants. TS6_{CCAA} and TS9_{CCAA} provide further evidence for the idea that the newly available e-learning tools were designed correctly and diverse enough to support everyone's expectations and preferences. This also aligns with performances in grades for the micro-challenges along the course of the term where despite the absence of statistically significant differences for three periods of observation (i.e., 2018-1, 2018-2, 2019-1), a tendency of improvement was observed (Fig. 5A). Moreover, according to the course's evaluation matrix, the average grades showed a proficiency for the student.

In the professional skills component, an improvement was observed for PS1_{CCAA} and PS9_{CCAA}. The results suggest that in the first place, students felt more comfortable with the number and pace at which activities were assigned throughout the term. Additionally, and in line with the Effects and Impacts Survey's findings regarding the course environment, students highly appreciated the possibility of having more open discussions with peers. As discussed above, this is critical to start promoting the social connectedness required to maintain students' engagement and, ultimately, long-lasting learning (Borup et al., 2020).

Our observations of superior student engagement with multimedia materials and tools support the idea that students are more prone to use ICTs as they familiarize themselves with them, which is valuable in preparation for exams and homework. Similar studies in different areas have demonstrated that students progressed toward more complex reasoning and expert conceptions as they master the tools in technology-mediated environments (Reilly et al., 2019). However, it is important to note that even though learning is facilitated as students improve their technology management skills, this is insufficient to create arguments or inferences (Wang et al., 2011). Engagement increased in time during the term, possibly as a response to the modules' complexity or the enhanced perception that knowledge can be gained from alternative sources, which are virtual in this case.

The need for improvement in TS1_{CCAA} suggests that despite professors and graduate teaching assistants' efforts to provide relevant feedback, additional work is needed to ensure that concerns come to everyone's full resolution. As discussed above, it is critical to focus on strategies to promote long-lasting relationships where students find an environment of cognitive development and emotional support.

An important aspect of improvement in the ICT framework is related to PS6_{CCAA}, which confirms that despite the general interest of students in ICTs, additional efforts need to be dedicated to developed extra online platforms for collaboration and teamwork. This also aligns well with the idea put forward above of more engaging communities of interaction for learning. A low score for PS14_{CCAA} indicates that both undergraduate tutors and graduate students fail to fully align with professors to promote an environment for peer-to-peer interactions that go beyond superficial discussions. As presented above, this is of the utmost importance for connectedness and setting up a baseline of shared knowledge that could eventually translate into higher-level thinking and constructive activities. According to the most recent tendencies in good practices for maintaining healthy teamwork environments, it is critical to assure that peer interactions are always under a frame of accountability and mutual respect (Cleary and Horsfall, 2015; Smith, 2000). Despite these results, students perceived themselves as fully involved in teamwork. A positive correlation

was found with the self- and co-evaluation grades. This is evidenced by the linear plot shown in Fig. 5B. The decrease in grades from 2017-2 to 2018-1 was attributed to the adaptation period after implementing just one module of the designed intervention.

The last group of items for improvement were TE8_{CCAA}, TE9_{CCAA}, and TE10_{CCAA}, strongly indicating that there are still some issues in assuring that students' contents and responses are posted opportunistically. This is a critical aspect of technology-based interventions' success, as that is one of the main attributes of the associated value proposition. A similar finding was obtained from the Effects and Impacts Survey, and to address it, we propose to have the tools in several platforms (and particularly mobile) such that notifications are easy to post, receive and track in time. For this reason, most of the successful e-learning platforms not only have web-based systems but can be launched from mobile apps without losing critical functionalities (Dagger et al., 2007; Motiwalla, 2007). A multiplatform system is likely to promote more continuous engagement with the course activities and the possibility that the course personnel responds more attentively.

6. Conclusions

The rapidly changing job marketplace is currently looking for engineering graduates with a more comprehensive skill collection as they need to tackle complex and multiparametric challenges. Traditional engineering education approaches have provided an incomplete repertoire of competencies as they have focused on the sole development of technical skills. Education innovators have started to address this significant hurdle by implementing active-learning pedagogical frameworks where students are at the center of the experience. With these emerging frameworks, engineering curricula worldwide have had profound transformations towards developing teamwork and people-related skills and the ability to operate in technology-enabled environments. Herein, we successfully implemented a framework that comprised three major components: a flipped-classroom approach to develop technical skills, a PO-PBL project to strengthen professional skills and e-learning tools to deepen students' technological expertise. This framework allowed us to increase the students' perception regarding the practical apprehension of concepts and the development of project management, negotiation, autonomy, communication, peer-to-peer interactions, and self-regulated skills.

Moreover, the proposed framework was vital to promote the use of e-books, interactive presentations, and video clips, which was reflected in an increased level of engagement as time progressed during the term. Although students evaluated most items with the highest scores, some aspects for improvement include a multiplatform availability of ICTs, a more attentive and prompter response by professors and teaching assistants, and the need for a more comfortable and positive environment for discussions when the teaching assistants are involved. Future interventions should most importantly focus on providing opportunities to students for creating communities of engagement mediated by technology where they can find spaces for building up more robust peer-to-peer connections. We are also planning to continue investigating the importance of genuine student involvement in teamwork as a major pillar for achieving performance above the average. Our results suggest that the developed framework can be extended to other engineering courses where interventions are required to ensure the comprehensive development of skills in accordance with quality education standards, international accreditation certificates, or curricular reforms.

Funding

This work was supported by the Department of Chemical and Food Engineering, and the School of Engineering, Universidad de los Andes, Bogota, Colombia.

CRedit authorship contribution statement

Miguel Ángel Ballesteros: Investigation, Writing - review & editing. **Juan Sebastián Sánchez:** Formal analysis, Writing - original draft, Writing - review & editing. **Nicolás Ratkovich:** Conceptualization, Methodology, Investigation, Writing - review & editing, Supervision, Project administration, Funding acquisition. **Juan C. Cruz:** Formal analysis, Writing - original draft, Supervision. **Luis H. Reyes:** Conceptualization, Methodology, Formal analysis, Investigation, Writing - original draft, Writing - review & editing, Visualization, Supervision, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

Acknowledgements

The authors would like to thank the Department of Chemical and Food Engineering and Conecta-TE (Center for Innovation in Technology and Education), especially Diana Cortés and Alexandra Ramirez, who provided insight and expertise for the course redesign. We would also like to show our gratitude to Miguel Bravo, Nicolás Pérez, Deisy Becerra, Laura Munar, Carolina Esquivel, David Santamaría, Joan Galindo, Daniela Bolaños, and Ximena Vargas for their assistance.

References

- Abidi, S.H., Pasha, A., Ali, S., 2016. Participant experience of the first massive open online course (MOOC) from Pakistan. *Curr. Issues Emerg. eLearning* 3, 11.
- Ahn, M.L., Yoon, H.S., Cha, H.J., 2015. Cultural sensitivity and design implications of MOOCs from Korean learners' perspectives: case studies on edX and coursera. *Educ. Technol. Int.* 16, 201–229.
- Alaeddine, N.I., Parsaei, H.R., Kakosimos, K., Guo, B., Mansoor, B., 2015. Teaching innovation with technology to accelerate engineering students' learning. ASEE Annual Conference and Exposition, Conference Proceedings, 24815, <http://dx.doi.org/10.18260/p.24815>.
- Backlund, G., Sjunnesson, J., 2012. Training young engineers to see. *AI Soc.* 27, 509–515, <http://dx.doi.org/10.1007/s00146-011-0368-9>.
- Ballesteros, M.A., Daza, M.A., Valdés, J.P., Ratkovich, N., Reyes, L.H., 2019. Applying PBL methodologies to the chemical engineering courses: unit operations and modeling and simulation, using a joint course project. *Educ. Chem. Eng.*, <http://dx.doi.org/10.1016/j.ece.2019.01.005>.
- Bates, T., 2001. The continuing evolution of ICT capacity: the implications for education. *Chang. Faces Virtual Educ.*, 29–46.
- Bauman, A., 2018. Concept maps: active learning assessment tool in a strategic management capstone class. *Coll. Teach.* 66, 213–221, <http://dx.doi.org/10.1080/87567555.2018.1501656>.
- Bernacki, M.L., Greene, J.A., Crompton, H., 2020. Mobile technology, learning, and achievement: advances in understanding and measuring the role of mobile technology in education. *Contemp. Educ. Psychol.* 60, 101827, <http://dx.doi.org/10.1016/j.cedpsych.2019.101827>.
- Borup, J., Graham, C.R., West, R.E., Archambault, L., Spring, K.J., 2020. Academic communities of engagement: an expansive lens for examining support structures in blended and online learning. *Educ. Technol. Res. Dev.* 68, 807–832, <http://dx.doi.org/10.1007/s11423-020-09744-x>.
- Bosman, L.B., Keller, J.C., Mentzer, N., Sparkling, A.E., 2020. Applying multiple modes of assessment to evaluate the team work competence. *Int. J. Eng. Educ.* 36, 317–327.
- Boud, D., Soler, R., 2016. Sustainable assessment revisited. *Assess. Eval. High. Educ.* 41, 400–413, <http://dx.doi.org/10.1080/02602938.2015.1018133>.
- Burgess, A., Haq, I., Bleasel, J., Roberts, C., Garsia, R., Randal, N., Mellis, C., 2019. Team-based learning (TBL): a community of practice. *BMC Med. Educ.* 19, 369.
- Cambini, C., Jiang, Y., 2009. Broadband investment and regulation: a literature review. *Telecomm. Policy* 33, 559–574, <http://dx.doi.org/10.1016/j.telpol.2009.08.007>.
- Carle, A.C., Jaffee, D., Miller, D., 2009. Engaging college science students and changing academic achievement with technology: a quasi-experimental preliminary investigation. *Comput. Educ.* 52, 376–380, <http://dx.doi.org/10.1016/j.compedu.2008.09.005>.
- Chen, J., Kolmos, A., Du, X., 2020. Forms of implementation and challenges of PBL in engineering education: a review of literature. *Eur. J. Eng. Educ.*, 1–26, <http://dx.doi.org/10.1080/03043797.2020.1718615>.
- Cleary, M., Horsfall, J., 2015. Teamwork and teambuilding: considering retreats. *Issues Ment. Health Nurs.* 36, 78–80, <http://dx.doi.org/10.3109/01612840.2014.981432>.
- Crumly, C., Dietz, P., d'Angelo, S., 2017. Pedagogies for Student-Centered Learning, Pedagogies for Student-Centered Learning. Augsburg Fortress Publishers, <http://dx.doi.org/10.2307/j.ctt9m0skc>.
- Dagger, D., O'Connor, A., Lawless, S., Walsh, E., Wade, V.P., 2007. Service-oriented e-learning platforms: from monolithic systems to flexible services. *IEEE Internet Comput.* 11, 28–35, <http://dx.doi.org/10.1109/MIC.2007.70>.
- Donnelly, R., Fitzmaurice, M., 2014. Collaborative project-based learning and problem-based learning in higher education: a consideration of tutor and students roles. *Emerg. Issues Pract. Univ. Learn. Teach.*, 87–98.
- Doo, M.Y., Bonk, C.J., 2020. The effects of self-efficacy, self-regulation and social presence on learning engagement in a large university class using flipped Learning. *J. Comput. Assist. Learn.*, <http://dx.doi.org/10.1111/jcal.12455>.
- Duhigg, C., 2016. What Google learned from its quest to build the perfect team [WWW Document]. *N. Y. Times Mag.*, <http://dx.doi.org/10.1017/CBO9781107415324.004>.
- Edmondson, A., 1999. Psychological safety and learning behavior in work teams. *Adm. Sci. Q.* 44, 350–383, <http://dx.doi.org/10.2307/2666999>.
- Edmondson, A.C., Lei, Z., 2014. Psychological safety: the history, renaissance, and future of an interpersonal construct. *Annu. Rev. Organ. Psychol. Organ. Behav.* 1, 23–43, <http://dx.doi.org/10.1146/annurev-orgpsych-031413-091305>.
- Edmondson, A.C., Kramer, R.M., Cook, K.S., 2004. Psychological safety, trust, and learning in organizations: a group-level lens. *Trust distrust Organ. Dilemmas Approaches* 12, 239–272.
- Edmunds, R., Thorpe, M., Conole, G., 2012. Student attitudes towards and use of ICT in course study, work and social activity: a technology acceptance model approach. *Br. J. Educ. Technol.* 43, 71–84, <http://dx.doi.org/10.1111/j.1467-8535.2010.01142.x>.
- Eom, S.B., Ashill, N.J., 2018. A system's view of E-Learning success model. *Decis. Sci. J. Innov. Educ.* 16, 42–76, <http://dx.doi.org/10.1111/dsji.12144>.
- Estrada, Á.C.M., Vera, J.G., Ruiz, G.R., Arrebola, I.A., 2019. Flipped classroom to improve university student centered learning and academic performance. *Soc. Sci.* 8, 315, <http://dx.doi.org/10.3390/socsci8110315>.
- Falconer, J.L., 2016. Why not try active learning? *AIChE J.* 62, 4174–4181, <http://dx.doi.org/10.1002/aic.15387>.
- Falconer, J.L., Nicodemus, G.D., Degrazia, J., Will Medlin, J., 2012. Chemical engineering screencasts. *Chem. Eng. Educ.* 46, 58–62.
- Fredricks, J.A., Blumenfeld, P., Friedel, J., Paris, A., 2006. School engagement. In: *What Do Children Need to Flourish?* Springer, pp. 305–321, http://dx.doi.org/10.1007/0-387-23823-9_19.
- Galvis, A.H., 1992. Teorías de aprendizaje como sustento al diseño de ambientes de aprendizaje. In: *Ingeniería de Software Educativo*. Ediciones Uniandes, pp. 85–123.
- Galvis, A.H., Galvis, F.A., Casas, J.P., Valencia, D.M., Avalo, A., Jiménez, A.T., 2017. Rethinking engineering education practices with active learning and digital technologies: pilot experiences at the University of the Andes, Bogotá. In: *2017 Research in Engineering Education Symposium, REES 2017, REES 2017*.
- Goldwasser, M., Mankoff, C., Manturuk, K., Schmid, L., Whitfield, K.E., 2016. Who is a student: completion in coursera courses at duke university. *Curr. Issues Emerg. eLearning* 3, 125.
- Gong, D., Yang, H.H., Cai, J., 2020. Exploring the key influencing factors on college students' computational thinking skills through flipped-classroom instruction. *Int. J. Educ. Technol. High. Educ.* 17, 1–13, <http://dx.doi.org/10.1186/s41239-020-00196-0>.
- Grasso, D., Burkins, M.B., 2010. Holistic Engineering Education: Beyond Technology, Holistic Engineering Education: Beyond Technology. Springer Science & Business Media, <http://dx.doi.org/10.1007/978-1-4419-1393-7>.
- Haidet, P., Kubitz, K., McCormack, W.T., 2014. Analysis of the team-based learning literature: TBL comes of age. *J. Excell. Coll. Teach.* 25, 303–333.
- Handley, K., Sturdy, A., Fincham, R., Clark, T., 2006. Within and beyond communities of practice: making sense of learning through participation, identity and practice. *J. Manag. Stud.* 43, 641–653, <http://dx.doi.org/10.1111/j.1467-6486.2006.00605.x>.
- Hasna, A.M., 2010. E-competence in chemical engineering learning and teaching. In: *ICETC 2010 - 2010 2nd International Conference on Education Technology and Computer*, IEEE, pp. V4–330, <http://dx.doi.org/10.1109/ICETC.2010.5529669>.
- Henrie, C.R., Halverson, L.R., Graham, C.R., 2015. Measuring student engagement in technology-mediated learning: a review. *Comput. Educ.* 90, 36–53, <http://dx.doi.org/10.1016/j.compedu.2015.09.005>.
- Hinds, P.J., Bailey, D.E., 2003. Out of sight, out of sync: understanding conflict in distributed teams. *Organ. Sci.* 14, 615–632, <http://dx.doi.org/10.1287/orsc.14.6.615.24872>.

- Hollingworth, R.W., McLoughlin, C., 2001. Developing science students' metacognitive problem solving skills online. *Australas. J. Educ. Technol.* 17, <http://dx.doi.org/10.14742/ajet.1772>.
- Hsu, Y.C., Ching, Y.H., 2013. Mobile computer-supported collaborative learning: a review of experimental research. *Br. J. Educ. Technol.* 44, <http://dx.doi.org/10.1111/bjet.12002>.
- Hutchings, M., Quinney, A., 2015. The flipped classroom, disruptive pedagogies, enabling technologies and wicked problems: responding to 'the bomb in the basement'. *Electron. J. e-Learning* 13, 105–118.
- Jeng, Y.L., Wu, T.T., Huang, Y.M., Tan, Q., Yang, S.J.H., 2010. The add-on impact of mobile applications in learning strategies: a review study. *Educ. Technol. Soc.* 13, 3–11.
- Jones, B.F., Rasmussen, C.M., Moffitt, M.C., 2004. Real-life Problem Solving: a Collaborative Approach to Interdisciplinary Learning. Real-life Problem Solving: a Collaborative Approach to Interdisciplinary Learning. American Psychological Association, <http://dx.doi.org/10.1037/10266-000>.
- Kilic, D., 2016. An examination of using self-, peer-, and teacher-assessment in higher education: a case study in teacher education. *High. Educ. Stud.* 6, 136, <http://dx.doi.org/10.5539/hes.v6n1p136>.
- King, A., 2002. Structuring peer interaction to promote high-level cognitive processing. *Theory Pract.* 41, 33–39, <http://dx.doi.org/10.1207/s15430421tip4101.6>.
- King, J.E., 2009. Educating engineers for the 21st century. *London R. Acad. Eng.* 3–28, http://dx.doi.org/10.1142/9789814261784_0001.
- Kokotsaki, D., Menzies, V., Wiggins, A., 2016. Project-based learning: a review of the literature. *Improv. Sch.* 19, 267–277.
- Kolowich, S., 2013. How EdX plans to earn, and share, revenue from free online courses. *Chron. High. Educ.* 59, A8.
- Låg, T., Sæle, R.G., 2019. Does the flipped classroom improve student learning and satisfaction? A systematic review and meta-analysis. *AERA Open* 5, 233285841987048, <http://dx.doi.org/10.1177/2332858419870489>.
- Lam, R., 2017. Enacting feedback utilization from a task-specific perspective. *Curric. J.* 28, 266–282, <http://dx.doi.org/10.1080/09585176.2016.1187185>.
- Leask, M., 2012. Issues in Teaching Using ICT. Issues in Teaching Using ICT. Psychology Press, <http://dx.doi.org/10.4324/9780203185117>.
- Lee, D., Huh, Y., Reigeluth, C.M., 2015. Collaboration, intragroup conflict, and social skills in project-based learning. *Instr. Sci.* 43, 561–590.
- Leenknecht, M.J.M., Wijnia, L., Loyens, S.M.M., Rikers, R.M.J.P., 2017. Need-supportive teaching in higher education: configurations of autonomy support, structure, and involvement. *Teach. Teach. Educ.* 68, 134–142, <http://dx.doi.org/10.1016/j.tate.2017.08.020>.
- Levasseur, R.E., 2017. People skills: building the perfect team—a change management perspective. *Interfaces (Providence)* 47, 270–272, <http://dx.doi.org/10.1287/inte.2017.0896>.
- Liberatore, M.W., Marr, D.W.M., Herring, A.M., Way, J.D., 2013. Student-created homework problems: based on youtube videos. *Chem. Eng. Educ.* 47, 122–132.
- Lofstrom, E., Nevgi, A., 2008. University teaching staffs' pedagogical awareness displayed through ICT-facilitated teaching. *Interact. Learn. Environ.* 16, 101–116, <http://dx.doi.org/10.1080/10494820701282447>.
- Lombardi, M.M., Oblinger, D.G., 2007. Authentic learning for the 21st century: an overview. *Learning* 1, 1–7.
- MacLeod, J., Yang, H.H., Zhu, S., Shi, Y., 2018. Technological factors and student-to-student connected classroom climate in cloud classrooms. *J. Educ. Comput. Res.* 56, 826–847, <http://dx.doi.org/10.1177/0735633117733999>.
- MacLeod, J., Yang, H.H., Shi, Y., 2019. Student-to-student connectedness in higher education: a systematic literature review. *J. Comput. High. Educ.* 31, 426–448, <http://dx.doi.org/10.1007/s12528-019-09214-1>.
- Melior, M.Y., Maimun, A.L., Chua, P.L., 2009. Language learning via ICT: uses, challenges and issues. *WSEAS Trans. Inf. Sci. Appl.* 6, 1453–1467.
- Menekse, M., Chi, M.T.H., 2019. The role of collaborative interactions versus individual construction on students' learning of engineering concepts. *Eur. J. Eng. Educ.* 44, 702–725, <http://dx.doi.org/10.1080/03043797.2018.1538324>.
- Mercader, C., Ion, G., Díaz-Vicario, A., 2020. Factors influencing students' peer feedback uptake: instructional design matters. *Assess. Eval. High. Educ.*, 1–12, <http://dx.doi.org/10.1080/02602938.2020.1726283>.
- Mergendoller, J.R., Markham, T., Ravitz, J., Larmer, J., 2015. Pervasive management of project based learning. *Handb. Classr. Manag.*, http://dx.doi.org/10.4324/9780203874783_ch22.
- Michalsky, T., Schechter, C., 2013. Preservice teachers' capacity to teach self-regulated learning: integrating learning from problems and learning from successes. *Teach. Teach. Educ.* 30, 60–73, <http://dx.doi.org/10.1016/j.tate.2012.10.009>.
- Mio, C., Ventura-Medina, E., João, E., 2019. Scenario-based eLearning to promote active learning in large cohorts: Students' perspective. *Comput. Appl. Eng. Educ.* 27, 894–909, <http://dx.doi.org/10.1002/cae.22123>.
- Moore, G.T., Piaget, J., 1971. Science of education and the psychology of the child. *J. Archit. Educ.* 25, 113, <http://dx.doi.org/10.2307/1423801>.
- Motiwalla, L.F., 2007. Mobile learning: a framework and evaluation. *Comput. Educ.* 49, 581–596, <http://dx.doi.org/10.1016/j.compedu.2005.10.011>.
- Musa, F., Mufti, N., Latiff, R.A., Amin, M.M., 2012. Project-based learning (PjBL): inculcating Soft skills in 21st century workplace. *Proc. - Soc. Behav. Sci.* 59, 565–573, <http://dx.doi.org/10.1016/j.sbspro.2012.09.315>.
- Ndoye, A., 2017. Peer/self assessment and student learning. *Int. J. Teach. Learn. High. Educ.* 29, 255–269.
- Ng, E.M.W., 2018. Integrating self-regulation principles with flipped classroom pedagogy for first year university students. *Comput. Educ.* 126, 65–74, <http://dx.doi.org/10.1016/j.compedu.2018.07.002>.
- Organisation for Economic Co-operation and Development, 2015. The future of productivity. *Jt. Econ. Dep. Dir. Sci. Technol. Innov. Policy Note*.
- Organisation for Economic Co-operation and Development, 2016. Skills for a Digital World, Policy Brief on The Future of Work.
- Parappilly, M., Woodman, R.J., Randhawa, S., 2019. Feasibility and effectiveness of different models of team-based learning approaches in STEMM-Based disciplines. *Res. Sci. Educ.*, 1–15, <http://dx.doi.org/10.1007/s11665-019-09888-8>.
- Puntambekar, S., 2006. Analyzing collaborative interactions: divergence, shared understanding and construction of knowledge. *Comput. Educ.* 47, 332–351, <http://dx.doi.org/10.1016/j.compedu.2004.10.012>.
- Rashid, U., Ventura-Medina, E., 2012. Scenario based e-learning to improve problem solving skills, in: *Profession of Engineering. ... Engineers Australia*, 238.
- Reeves, P.M., Sperling, R.A., 2015. A comparison of technologically mediated and face-to-face help-seeking sources. *Br. J. Educ. Psychol.* 85, 570–584, <http://dx.doi.org/10.1111/bjep.12088>.
- Reilly, C.M., Kang, S.Y., Grotzer, T.A., Joyal, J.A., Oriol, N.E., 2019. Pedagogical moves and student thinking in technology-mediated medical problem-based learning: supporting novice-expert shift. *Br. J. Educ. Technol.* 50, 2234–2250, <http://dx.doi.org/10.1111/bjet.12843>.
- Resta, P., Laferrière, T., 2007. Technology in support of collaborative learning. *Educ. Psychol. Rev.* 19, 65–83, <http://dx.doi.org/10.1007/s10648-007-9042-7>.
- Richards, J.C., Burns, A., 2012. The Cambridge Guide to Pedagogy and Practice in Second Language Teaching. Cambridge University Press.
- Salcedo Galán, F., Buitrago Mora, H.M., Jiménez Heredia, A.T., Sierra Peñuela, L.L., González Barrios, A.F., 2018. Everyday chemical product design as platform for teaching transport phenomena. *Educ. Chem. Eng.* 25, 9–15, <http://dx.doi.org/10.1016/j.ece.2018.09.001>.
- Saunders, G., Klemming, F., 2003. Integrating technology into a traditional learning environment: reasons for and risks of success. *Act. Learn. High. Educ.* 4, 74–86, <http://dx.doi.org/10.1177/1469787403004001862>.
- Sein-Echaluze, M.L., Fidalgo-Blanco, Á., Esteban-Escañó, J., García-Peñalvo, F.J., Conde, M.Á., 2018. Using learning analytics to detect authentic leadership characteristics in engineering students. *Int. J. Eng. Educ.* 34, 851–864.
- Shi, Y., Yang, H., MacLeod, J., Zhang, J., Yang, H.H., 2020. College students' cognitive learning outcomes in technology-enabled active learning environments: a meta-analysis of the empirical literature. *J. Educ. Comput. Res.* 58, 791–817, <http://dx.doi.org/10.1177/0735633119881477>.
- Sidelinger, R.J., Bolen, D.M., McMullen, A.L., Nyeste, M.C., 2015. Academic and social integration in the basic communication course: predictors of students' out-of-Class communication and academic learning. *Commun. Stud.* 66, 63–84, <http://dx.doi.org/10.1080/10510974.2013.856807>.
- Smith, K.A., 2000. Strategies for developing engineering student's teamwork and project management skills. *ASEE Annual Conference Proceedings*, 5277–5288.
- Song, D., Rice, M., Oh, E.Y., 2019. Participation in online courses and interaction with a virtual agent. *Int. Rev. Res. Open Distance Learn* 20, 44–62, <http://dx.doi.org/10.19173/irrodl.v20i1.3998>.
- Tong, Lee Fui, 2003. Identifying essential learning skills in students' Engineering education, Monash University Malaysia. *Proc. HERDSA* 2003.
- Valentín, A., Mateos, P.M., González-Tablas, M.M., Pérez, L., López, E., García, I., 2013. Motivation and learning strategies in the use of ICTs among university students. *Comput. Educ.* 61, 52–58, <http://dx.doi.org/10.1016/j.compedu.2012.09.008>.
- Vasquez, E.S., Dewitt, M.J., West, Z.J., Elsass, M.J., 2020. Impact of team formation approach on teamwork effectiveness and performance in an upper-level undergraduate chemical engineering laboratory course. *Int. J. Eng. Educ.* 36, 491–501.
- Voogt, J., Pelgrum, H., 2005. ICT and curriculum change. *Hum. Technol. An Interdiscip. J. Humans ICT Environ.* 1, 157–175, <http://dx.doi.org/10.17011/ht/urn.2005356>.
- Vovides, Y., Sanchez-Alonso, S., Mitropoulou, V., Nickmans, G., 2007. The use of e-learning course management systems to support learning strategies and to improve self-regulated learning. *Educ. Res. Rev.* 2, 64–74, <http://dx.doi.org/10.1016/j.edurev.2007.02.004>.
- Vygotsky, L.S., 1980. *Mind in Society: the Development of Higher Psychological Processes*. Harvard university press.
- Wall, K., 2010. Engineering: Issues, Challenges and Opportunities for Development. UNESCO.
- Wang, M., Peng, J., Cheng, B., Zhou, H., Liu, J., 2011. Knowledge visualization for self-regulated learning. *Educ. Technol. Soc.* 14, 28–42.
- Wentzel, K.R., Watkins, D.E., 2011. Peer relationships and learning: implications for instruction. *Handb. Res. Learn. Instr.*, 322–343.
- Wong, E.M.L., Li, S.C., 2011. Framing ICT implementation in a context of educational change: a structural equation modelling analysis. *Australas. J. Educ. Technol.* 27, 361–379, <http://dx.doi.org/10.14742/ajet.975>.
- Woods, D.R., 2012. PBL: an evaluation of the effectiveness of authentic problem-based learning (aPBL). *Chem. Eng. Educ.* 46, 135–144.
- Woods, D.R., Sheardown, H., 2009. Ideas for creating and overcoming student silences. *Chem. Eng. Educ.* 43, 125–130.
- Woods, D., Marshall, R.R., Hrymak, A.N., 1988. Self assessment in the context of the mcmaster problem solving programme. *Assess. Eval. High. Educ.* 13, 107–127, <http://dx.doi.org/10.1080/0260293880130203>.
- Woods, D.R., Hrymak, A.N., Marshall, R.R., Wood, P.E., Crowe, C.M., Hoffman, T.W., Wright, J.D., Taylor, P.A., Woodhouse, K.A., Bouchard, C.G.K., 1997. Developing

- problem solving skills: the McMaster problem solving program. *J. Eng. Educ.* 86, 75–91, <http://dx.doi.org/10.1002/j.2168-9830.1997.tb00270.x>.
- Woods, D.R., Briedis, D., Perna, A., 2013. Professional skills needed by our graduates. *Chem. Eng. Educ.* 47, 81–90.
- Woratschek, C.R., Lenox, T.L., 2002. Information systems entry-level job skills: a survey of employers. *Information Systems Educators Conference*, Vol. 19.
- World Economic Forum, 2017. Global shapers annual survey. *World Econ. Forum* 74, 71–74, <http://dx.doi.org/10.1177/003754970007400201>.
- Xu, Z., Shi, Y., 2018. Application of constructivist theory in flipped classroom – take college english teaching as a case study. *Theory Pract. Lang. Stud.* 8, 880, <http://dx.doi.org/10.17507/tpls.0807.21>.
- Yuen, K.M., Hau, K.T., 2006. Constructivist teaching and teacher-centred teaching: a comparison of students' learning in a university course. *Innov. Educ. Teach. Int.* 43, 279–290, <http://dx.doi.org/10.1080/14703290600750861>.